# $R_0$ and stability: will an outbreak grow?

Module 2 • next-generation matrix, equilibria and the early growth rate (constant $\beta$)

**Next-generation matrix**: split the $(e, i)$ equations into new infections $F$ and moves between classes $V$

$$F = \begin{pmatrix} 0 & \beta s^* \\ 0 & 0 \end{pmatrix}, \quad V = \begin{pmatrix} \sigma & 0 \\ -\sigma & \gamma + \mu \end{pmatrix}$$

$$R_0 = \rho(FV^{-1}) \text{ at } s^* = 1 \;=\; \frac{\beta}{\gamma + \mu}$$

- rate × time: 0.486 /day × 4.1 infectious days ≈ **2.00**
- $\sigma$ drops out: incubation changes *when* people infect others, not *how many*
- over time: $R_t = \beta(t)\, s(t)/(\gamma + \mu)$, which the PINN gives us

**Early growth rate** = the positive eigenvalue

$\lambda_+ = 0.062734$ /day (theory), 0.062732 (simulation)

- doubling time **11.1 days**; $R_0 = (1 + \frac{\lambda_+}{\sigma})(1 + \frac{\lambda_+}{\gamma+\mu})$ gives back 2.0004

**Equilibria**: every one is disease-free, $(s^*, 0, 0, r^*, d^*)$

no births and lasting immunity, so there is no endemic state

| $s^*$ | $R_0 s^*$ | eigenvalues $(e, i$ block) | result |
|---|---|---|---|
| 1.00 | 2.00 | $-0.412,\ +\mathbf{0.063}$ | unstable |
| 0.75 | 1.50 | $-0.383,\ +\mathbf{0.034}$ | unstable |
| 0.40 | 0.80 | $-0.334,\ -0.016$ | stable |
| 0.25 | 0.50 | $-0.307,\ -0.042$ | stable |

- threshold $R_0 s^* = 1$: herd immunity once $s^* < 1/R_0 = 50\%$
- **stable, not asymptotically**: three eigenvalues are 0, so the system settles at a *nearby* disease-free state

**One number decides growth:** a few infections grow if $R_0 s^* > 1$ and die out if $R_0 s^* < 1$. The growth rate in early case data gives $R_0$ back, and our simulation matches the theory to $3 \times 10^{-5}$.

# Final size: how many people get infected in total?

Module 2 • an exact answer without simulating, and a test for every numerical method in the project

**Divide $\dot{s}$ by $\dot{r} + \dot{d}$: $i$ cancels**

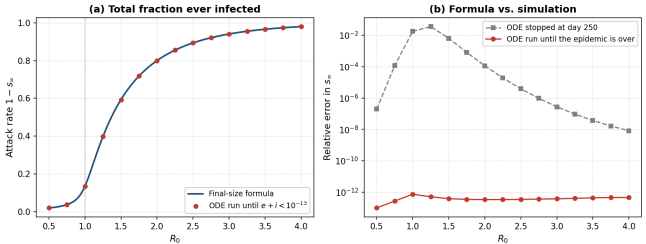
$$\frac{ds}{d(r+d)} = \frac{-\beta s i}{(\gamma+\mu)\, i} = -R_0\, s$$

**Integrate to the end of the epidemic**

$$\ln \frac{s_0}{s_\infty} = R_0\,(1 - s_\infty - r_0 - d_0)$$

$$d_\infty = d_0 + \underbrace{\frac{\mu}{\gamma+\mu}}_{\text{CFR } 41\%}\,(1 - s_\infty - r_0 - d_0)$$

- $R_0 = 2$: **80%** infected by the end, **33%** die (no interventions)
- exactly one root in $(0, s_0)$; found with Brent's method



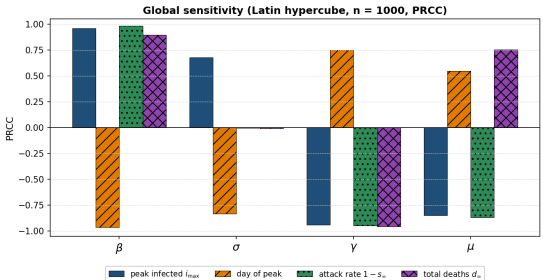

(a) formula (line) and ODE (dots); (b) error: run to the end (red) vs. stop at day 250 (grey)

- ✓ run the ODE until $e + i < 10^{-13}$: formula and ODE agree to $\mathbf{10^{-12}}$ for $R_0$ from 0.8 to 3.5
- × stop at day 250: errors up to **3.6%** near $R_0 = 1.25$, because slow epidemics last up to ∼1000 days

**The catch:** the $\beta$ cancels only if it is constant. With $\beta(t)$ there is no closed formula, which is one reason we need a data-driven method such as the PINN.

# Sensitivity: which parameters matter most?

Module 2 • Latin hypercube sampling ($n = 1000$, seed 42) and partial rank correlation (PRCC)




PRCC = one parameter's effect with the others held fixed

| | peak | peak day | attack rate | deaths |
|---|---|---|---|---|
| $\beta$ | $+0.96$ | $-0.96$ | $+0.98$ | $+0.90$ |
| $\sigma$ | $+0.68$ | $-0.83$ | $0.00^{\text{n.s.}}$ | $-0.01^{\text{n.s.}}$ |
| $\gamma$ | $-0.94$ | $+0.75$ | $-0.95$ | $-0.95$ |
| $\mu$ | $-0.85$ | $+0.55$ | $-0.87$ | $+0.75$ |

n.s. not significant at the 1% level.

Local index of $R_0$ (% change per 1%): $\beta$ +1, $\gamma$ −0.59, $\mu$ −0.41, $\sigma$ 0.

- $\beta$ raises everything: the main lever for control
- $\gamma$, $\mu$ shrink the epidemic, but $\mu$ **raises** deaths
- $\sigma$ moves the peak but not the final size, **as the formula predicts**

**What this means for the PINN:** cases and deaths alone cannot separate $\sigma$, $\gamma$ and $\mu$. So we fix $\sigma$ and $\gamma$ from clinical data; the PINN learns $\beta(t)$, and $\mu$ from the share of cases that die.